



Huang-Lian-Jie-Du extract ameliorates atopic dermatitis-like skin lesions induced by 2,4-dinitrobenzene in mice via suppression of MAPKs and NF- κ B pathways

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ABSTRACT

Ethnopharmacological relevance: Huang-Lian-Jie-Du Decoction (HLJDD), is a well-known traditional Chinese herbal formula first written in the Tang dynasty. In Chinese medicine practice, HLJDD is commonly prescribed to treat various inflammatory skin diseases, such as atopic dermatitis (AD) and psoriasis.

Aim of the study: The present study aimed at investigating the therapeutic effect of HLJDE extract (HLJDE) and to elucidate the underlying molecular mechanisms of action in the 1-chloro-2,4-dinitrobenzene (DNCB)-induced AD-like mice.

Materials and methods: Female Balb/c mice were sensitized with DNCB for three days. After sensitization, mice were challenged with DNCB every three days and orally administrated with HLJDE (150, 300 and 600 mg/kg) daily from day 14 to day 29 for consecutive 16 days. At the end of experiment, the clinical AD scores of the mice were calculated to evaluate the therapeutic effect of HLJDE, and serum, ears and dorsal skin of the mice were collected for unravelling molecular mechanisms.

Results: HLJDE significantly reduced the clinical symptoms in the AD-like mice by inhibiting eosinophil and mast cell infiltration, suppressing the production of Th2-associated cytokine (IL-4) and pro-inflammatory cytokines (TNF- α). In addition, HLJDE significantly suppressed the NF- κ B and MAPKs pathways. Moreover, HLJDE was able to accentuate filaggrin expression in the skin lesion when compared to the sensitized mouse without treatment.

Conclusion: HLJDE significantly improved the AD-like symptoms on the DNCB-sensitized mice through mitigating the production of inflammatory mediators via suppressing MAPKs and NF- κ B pathways. Additionally, the elevated expression of filaggrin in the skin lesion by HLJDE contributes to the recovery of dysfunctional skin barrier on the DNCB-sensitized mice.

1. Introduction

Atopic dermatitis (AD) is a common relapsing chronic inflammatory skin disease which has high prevalence all over the world, with 5–30% in children and 2–10% in adults. Notable clinical manifestations in AD patients include erythema, pruritus, excoriation, edema, exodus,

crusting and lichenification (Leung et al., 2004; Williams and Flohr, 2006). The immediate impact from this skin condition may have long-term negative impacts on academic performance, career achievement, mental stability and social activities. Moreover, AD is known to render substantial financial burdens to the patients and the health care systems as a whole. It has been reported that the annual cost for AD treatment

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Abbreviation

AD	Atopic dermatitis
CAM	Complementary and alternative medicine
DCs	Dendritic cells
DNCB	1-chloro-2,4-dinitrobenzene
DXM	Dexamethasone
FLG	Filaggrin null mutations

HLJDD	Huang-Lian-Jie-Du Decoction
IL-4	Interleukin-4
SDS-PAGE	Sodium dodecyl sulfate polyacrylamide gel electrophoresis
TCM	Traditional Chinese medicine
TNF- α	Tumor Necrosis Factor-alpha
TSLP	Thymic stromal lymphopoietin

may range from 71 to 2559 dollars per patient (Baptista et al., 2013).

AD is an allergic inflammatory disorder with a complex etiology, and several possible factors are believed to be involved in AD, including susceptibility genes, social or environmental factors, skin barrier dysfunction, and systemic and local immunologic responses. As an atopic skin disease, AD is also regarded as the cutaneous manifestation of a systemic disorder associated with an array of allergic diseases such as asthma, food allergy, and allergic rhinitis (Leung et al., 2004; Zheng et al., 2011). On the other hand, dysfunctional skin barrier directly increases the penetration of skin, rendering the host susceptible to various environmental insults, thereby aggravating the AD symptoms (Leung et al., 2004; Boguniewicz and Leung, 2011). Thus, the major clinical therapy of AD is through avoidance of various irritants and specific immunologic stimuli (foods and aeroallergens), protection of skin barrier, and symptom management usually involving long-term use of topical corticosteroids and calcineurin inhibitors, anti-inflammatory treatment, and UV light therapy (Leung et al., 2004; De Benedetto et al., 2012; Leung, 2000, 2013). Unfortunately, these therapies face with grim challenges, such as high cost in phototherapy, serious side effects and drug tolerance in corticosteroid and antibiotic therapy (Tan et al., 2013). For these reasons, patients frequently seek complementary and alternative medicine (CAM) as a therapeutic option for AD. Traditional Chinese medicine (TCM), including Chinese herbal medicine and acupuncture, remains one of the most popular forms of CAM (Bielory, 2002).

Huang-Lian-Jie-Du Decoction (HLJDD), a well-known traditional Chinese herbal formula first recorded in a medical book named *Arcane Essentials* from the Imperial Library (*Wai-Tai-Mi-Yao* in Chinese) written in the Tang dynasty, is composed of 4 herbs including the stem of *Coptis chinensis* Franch (*Coptidis Rhizoma*, Huang-Lian), the root of *Scutellaria baicalensis* Georgi (*Scutellariae Radix*, Huang-Qin), the bark of *Phellodendron amurense* Rupr. (*Phellodendri Chinensis Cortex*, Huang-Bai) and the fruit of *Gardenia jasminoides* Ellis (*Gardenia Fructus*, Zhi-Zi) at the ratio of 3:2:2:3 (Zeng et al., 2009). In Chinese medicine practice, HLJDD is known to possess therapeutic effects of purging fire and removing toxin, and is commonly prescribed to treat various skin diseases, such as AD and psoriasis (Chen et al., 2010; Ko and Baek, 2012). Besides these traditional actions and application, HLJDD has been demonstrated to possess various pharmacological activities, including anti-inflammatory (Wang and Xu, 2000), anti-oxidative (Zhang et al., 2016), anti-microbial (Xia et al., 2018), and anti-tumor actions (Lin et al., 2013).

In our previous study, we found that HLJDD exerted significant anti-inflammatory and anti-allergic effects through suppressing the production of allergic and inflammatory mediators via the NF- κ B and MAPKs inactivation and I κ B α degradation in the LPS-stimulated RAW24.7 cells, as well as inactivation of MAPKs and Lyn pathway in antigen-induced RBL-2H3 cells (Chen et al., 2016). These findings provide experimental evidence to support the clinical application of HLJDD for AD treatment. In the present study, we further affirmed the anti-AD effect of this ancient formula using the 1-chloro-2,4-dinitrobenzene (DNCB)-induced AD-like mouse model and elucidated the underlying molecular mechanisms of its action.

2. Materials and methods

2.1. Reagents and chemicals

Dexamethasone (DXM, purity $\geq$ 99%) and 1-chloro-2,4-dinitrobenzene (DNCB) were purchased from Sigma chemical Co. (St. Louis, MO, USA). All other reagents and chemicals used in the study were of analytical grade.

2.2. Animals

Female BALB/c mice (6–8 weeks old) were purchased from Laboratory Animal Services Centre, The Chinese University of Hong Kong, and housed in the animal room at the School of Chinese Medicine, Faculty of Medicine, CUHK. All mice were housed at constant temperature ($23 \pm 2^\circ\text{C}$) and humidity, and maintained on 12-h light and dark cycle. The experimental procedures were approved by the Animal Experimentation Ethics Committee of The Chinese University of Hong Kong (Ref. No. 16/112/MIS-6-B), and the animal license for this animal experiment was granted by the Department of Health, Hong Kong SAR.

2.3. Preparation of HLJDD extract

HLJDD extract (HLJDE) was prepared as previously described (Chen et al., 2016). Briefly, the crude decoction pieces of *Coptidis Rhizoma* (Huang-Lian), *Scutellariae Radix* (Huang-Qin), *Phellodendri Chinensis Cortex* (Huang-Bai) and *Gardenia Fructus* (Zhi-Zi) were purchased from Zhixin Herbal Pharmaceutical Company Ltd, a Guangzhou-based GMP accredited Chinese herbal supplier, and their identities were

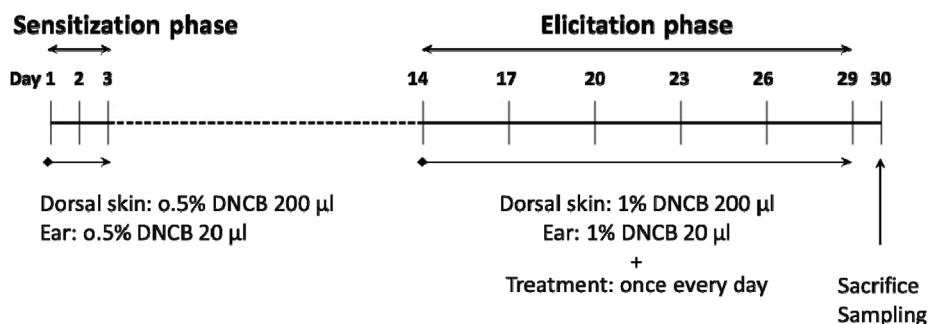


Fig. 1. Procedure of DNCB sensitization and challenge experiment in BALB/c mice. On day 1–3, the female mice were sensitized with 0.5% DNCB solution on the dorsal skin and the ears, followed by being challenged with 1% DNCB solution every three days from day 14 to day 29. The treatments with medications or 0.5% CMC-Na solution were conducted every day from day 14 to day 29. On day 30, the mice were sacrificed and sampled for further studies.

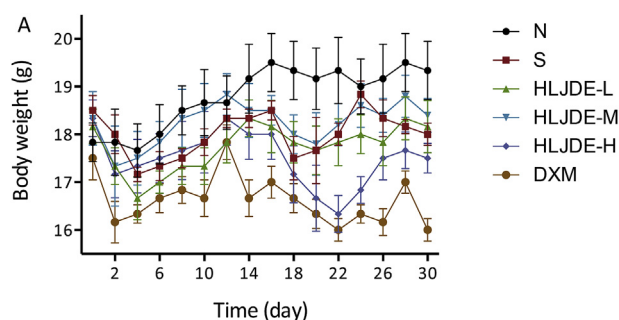


Fig. 2. Change in the body weight of the mice in the course of experiment. The body weights of mice were recorded at a regular time points. There was a total of 6 groups of mice used in this experiment: normal mice (N), sensitized mice (S), sensitized mice with low, medium and high dose of HLJDE (HLJDE-L, HLJDE-M and HLJDE-H), and sensitized mice with DXM (DXM). Data were mean $\pm$ SEM (n = 6 for each group).

authenticated in accordance with the Chinese Pharmacopoeia (2015 Edition). Voucher specimens of the above 4 herbs were deposited in the Herbarium of the School of Chinese Medicine, CUHK, with reference no. AD01–04, respectively. The above 4 herbs were ground to powder or pieces and mixed in the ratio 3:2:2:3, and the mixture was then extracted with 80% aqueous ethanol in an ultrasonic bath for 30 min. The extract was filtered and the residue was further extracted twice as before. All three filtrates were combined, and concentrated in a rotary evaporator under negative pressure, and finally dried in lyophilizer to get a freeze-dried power. The yield of HLJDE was 50.2%. The contents of five major components including geniposide, berberine, baicalin, wogonin and baicalin in the HLJDE were 6.13%, 5.94%, 0.76%, 0.29% and 3.62%, respectively; and a HPLC fingerprinting was constructed as a means of quality control for HLJDE as reported previously (Chen et al., 2016). The resultant HLHDE was store at -20°C for subsequent study.

2.4. Sensitization, challenge and treatment with the extracts

DNCB was dissolved in acetone/olive oil (3:1) to prepare proper

solutions before applied on mice. The procedure as depicted in Fig. 1 was performed as previously reported. Briefly, the mice were shaved under anesthesia (isoflurane) using electric shaver, and then 200 μL 0.5% DNCB or saline solution was applied to the shaved dorsal skin for 3 days. After 2 weeks, the mice were challenged with 200 μL 1% DNCB or saline solution on the same dorsal skin and also 20 μL 1% DNCB on each ear every 3 days, and this procedure was repeated for 6 times (on days 14, 17, 20, 23, 26, and 29). DNCB was applied on the same site of the skin and the ear.

A total of 6 testing groups (n = 6 per group) include 3 groups of HLJDE (150, 300, 600 mg/kg/day), normal control groups (challenged with normal saline), induced control group (sensitized and challenged with DNCB) and DXM group (2.5 mg/kg/day), as positive control group. After being challenged with 1% DNCB or saline, the mice were orally administrated with HLJDE, DXM or 0.5% CMC-Na once a day for 16 days.

2.5. Monitoring of the changes in body weight and clinical scores

The body weights of the mice were recorded every day. Before sacrifice, the mice were pictured for calculating the clinical scores according to the severity of symptoms on the dorsal skin and the ears. The main symptoms involved in clinical score include erythema, edema, crusts, excoriations, and lichenification.

2.6. Collection of serum and tissues

In the day after last DNCB challenge, the mice were anesthetized and the blood was harvested from the ophthalmic venous plexus, and the blood was centrifuged for 10 min at 2,000 rpm to collect serum for ELISA assay. The mice were sacrificed after blood collection and then the skin and the ears of mice were collected and stored at -80°C for further assays.

2.7. Histopathological studies

The samples of the skin and ears were fixed in 10% phosphate-buffered formalin. After two days, these tissues were dehydrated using

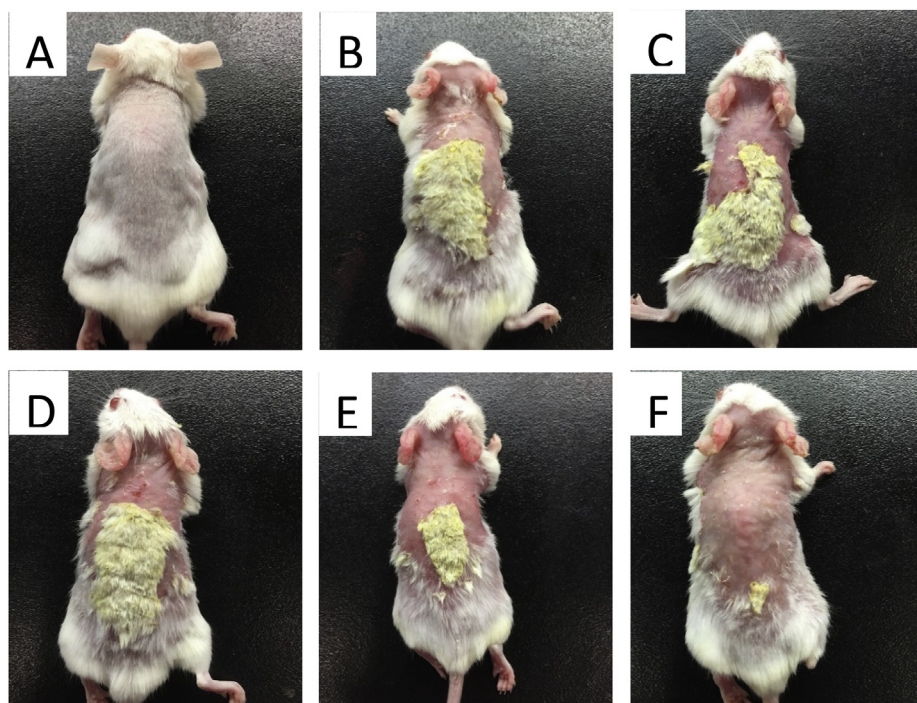


Fig. 3. Clinical features of AD-like lesions in BALB/c mice (entire skin). (A) normal mice, (B) sensitized mice, (C) sensitized mice with DXM, (D–F) sensitized mice with low, medium and high dose of HLJDE. n = 6 for all groups. Clinical features of the skin lesions in mice included erythema, edema, crusts, excoriations, and lichenification.

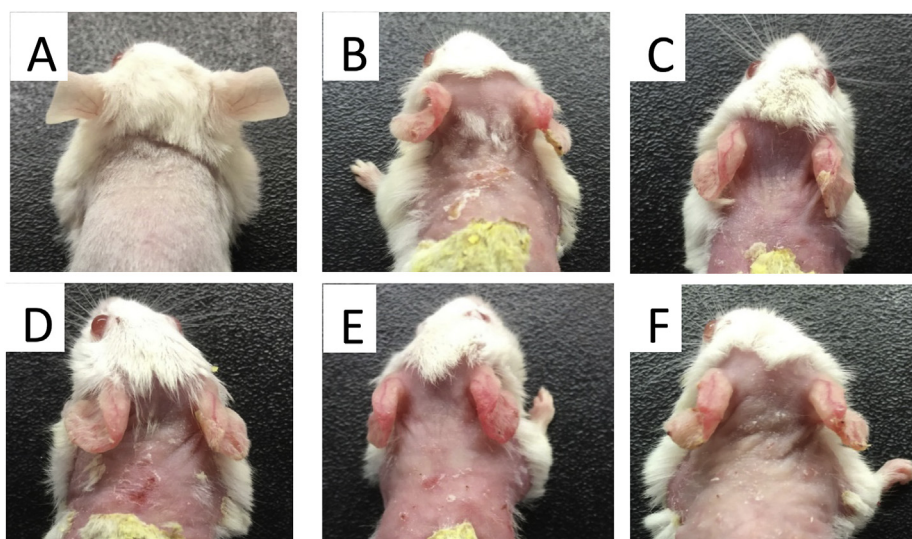


Fig. 4. Clinical features of AD-like lesions in BALB/c mice (Close-up photos). (A) normal mice, (B) sensitized mice, (C) sensitized mice with DXM, (D–F) sensitized mice with low, medium and high dose of HLJDE. $n = 6$ for all groups. Clinical features of skin lesions in mice included erythema, edema, crusts, excoriations and lichenification.

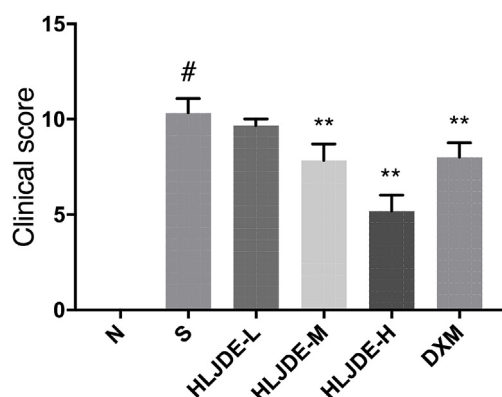


Fig. 5. Clinical scores of AD-like features in mice. The total clinical severity score for AD-like lesion was evaluated according to five main symptoms, including erythema, edema, crusts, excoriations and lichenification, with each symptom severity being graded as 0 (no symptoms), 1 (mild), 2 (intermediate), and 3 (severe). Data were mean $\pm$ SEM ($n = 6$). # $p < 0.01$ vs. normal mice (N); * $p < 0.05$, ** $p < 0.01$ vs. sensitized mice (S).

ethanol at different concentrations (from 70% to 100%) and subsequently embedded in paraffin. The tissues were sliced into 6 μ m sections and dried at 42 °C overnight for staining. The tissue sections were stained with hematoxylin and eosin (H & E) and toluidine blue after deparaffination and rehydration as previously described (Li et al., 2014). Briefly, for H & E staining, the sections were stained with hematoxylin for 5 min firstly, and stained with eosin for 5 min before washing with acidic alcohol. For toluidine blue staining, the sections were stained with toluidine blue solution for 3–5 min and washed with distilled water. Each section was dehydrated with ethanol at different concentrations (from 70% to 100%) and cleared with xylene, finally mounted on slide using mounting medium. The slides were observed and measured in five high-power fields (HPFs) using a light microscope (Nikon, Japan) at 200 $\times$ magnification.

2.8. Determination of serum cytokines and serum IgE level

The serum was obtained from the blood of the mice by centrifugation and stored at -80 °C for ELISA assay. The ELISA kits specific for Interleukin-4 (IL-4), tumor necrosis factor- α (TNF- α), thymic stromal lymphopoietin (TSLP) and IgE were purchased from eBioscience (California, USA), while the ELISA kit for serum histamine was obtained from Abnova (Heidelberg, Germany).

2.9. Determination of cytokines in the dorsal skin of mice

The dorsal skins were thawed and homogenized in ice PBS (1:5, w/v). The mixtures were centrifuged for 15 min at 3,000 rpm and the supernatant was collected for ELISA assay as per the manufacturer's instructions.

2.10. Preparation of nuclear protein and whole cell protein lysates

For preparation of whole protein lysate, the samples of the dorsal skin were homogenized in RIPA lysis buffer (1:10, w/v) and extracted on ice for 30 min and then centrifuged at 12,000 rpm at 4 °C. The supernatant was collected as the whole cell lysate and stored at -80 °C.

Nuclear protein lysate was extracted using a method described in our previous publication (Chen et al., 2016). Briefly, the dorsal skin was homogenized in hypotonic buffer (1:10, w/v) and incubated for 10 min on ice. The mixtures were centrifuged at 12,000 rpm for 10 min. The nuclear pellet was suspended in extraction buffer (1:3, w/v) for 20 min on ice and centrifuged for 15 min at 12,000 rpm at 4 °C. The supernatant was collected as nuclear protein lysate and stored at -80 °C.

2.11. Western blot analysis

The quantity of protein was measured using protein assay dye reagent (Bio-Rad, USA). Equal amounts of protein of different samples were separated by 10% sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE), and then transferred to PVDF membranes. After blocking with 5% nonfat milk for 1 h, the PVDF membranes were incubated overnight with the primary antibody at 4 °C, followed by rinsing the blot with TBST 3 times for 5 min each. The PVDF membranes were then incubated with HRP-conjugated secondary antibody solution for 1 h at room temperature. The protein bands were visualized by an ECL method, and the chemiluminescent signals detected and recorded by exposure of the membrane to x-ray film (GE Healthcare, Little Chalfont, UK).

2.12. Statistical analysis

The data were expressed as mean $\pm$ SEM. Multiple group comparisons were performed using one-way ANOVA followed by Dunnett's test. GraphPad Prism software was used to perform the statistical analysis (Version 4.0; GraphPad Software, Inc., San Diego, CA). Differences were considered statistically significant at $p < 0.05$.

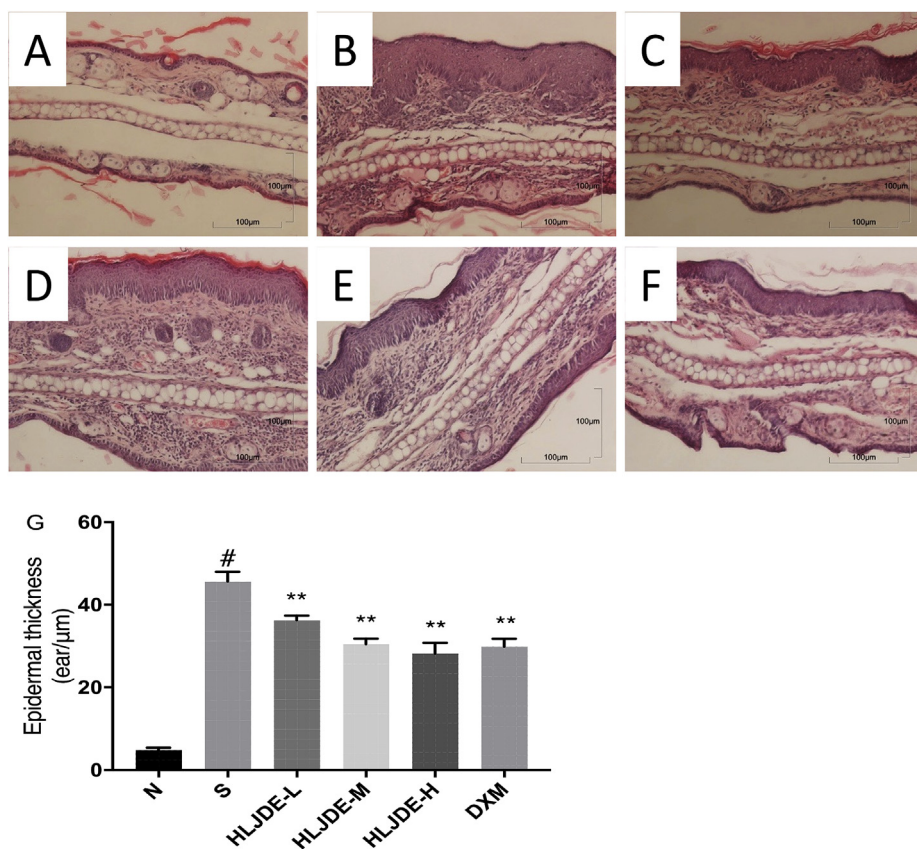


Fig. 6. Effects of HLJDE on the epidermal thickness in the ears of the sensitized mice. Ear sections were stained with H&E and analyzed for eosinophil infiltration and epidermal thickness (scale bar = 100 μm). (A) normal mice, (B) sensitized mice, (C) sensitized mice with DXM, (D–F) sensitized mice with low, medium and high dose of HLJDE, (G) quantification of epidermal thickness. Data were mean ± SEM (n = 6). #p < 0.01 vs. normal mice (N); *p < 0.05, **p < 0.01 vs. sensitized mice (S).

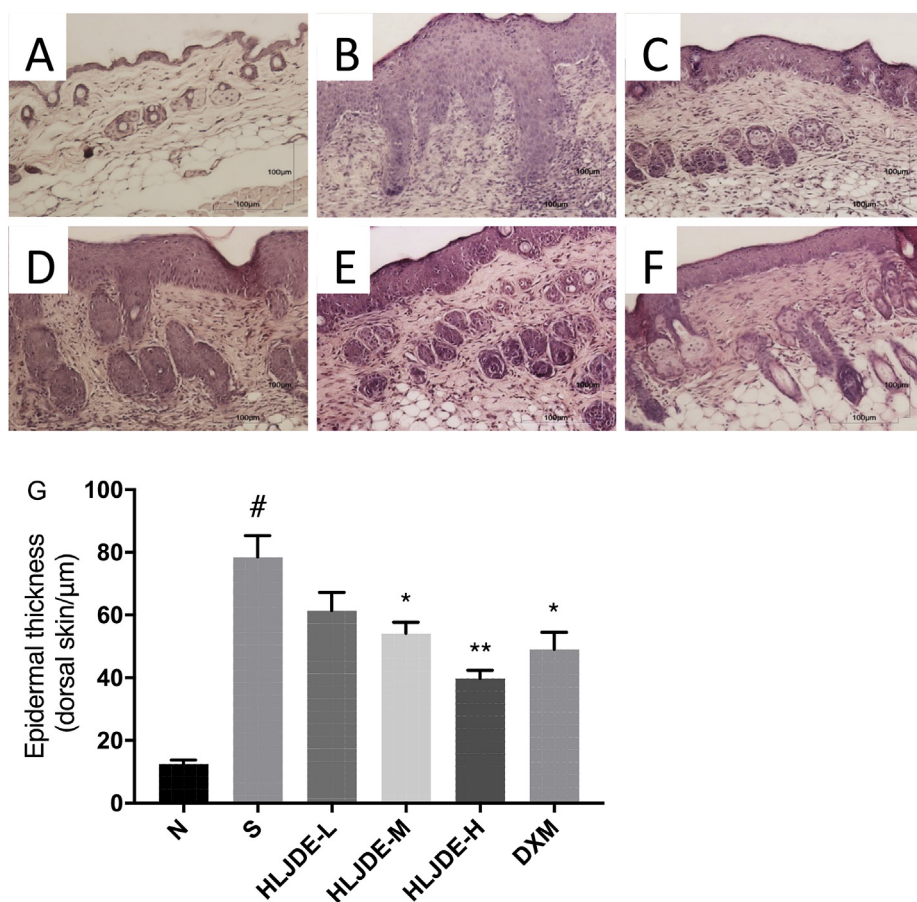


Fig. 7. Effects of HLJDD on the epidermal thickness in the dorsal skin. Dorsal skin sections were stained with H&E and analyzed for eosinophil infiltration and epidermal thickness (scale bar = 100 μm). (A) normal mice, (B) sensitized mice, (C) sensitized mice with DXM, (D–F) sensitized mice with low, medium and high dose of HLJDE, (G) epidermal thickness. Data were mean ± SEM (n = 6). #p < 0.01 vs. normal mice (N); *p < 0.05, **p < 0.01 vs. sensitized mice (S).

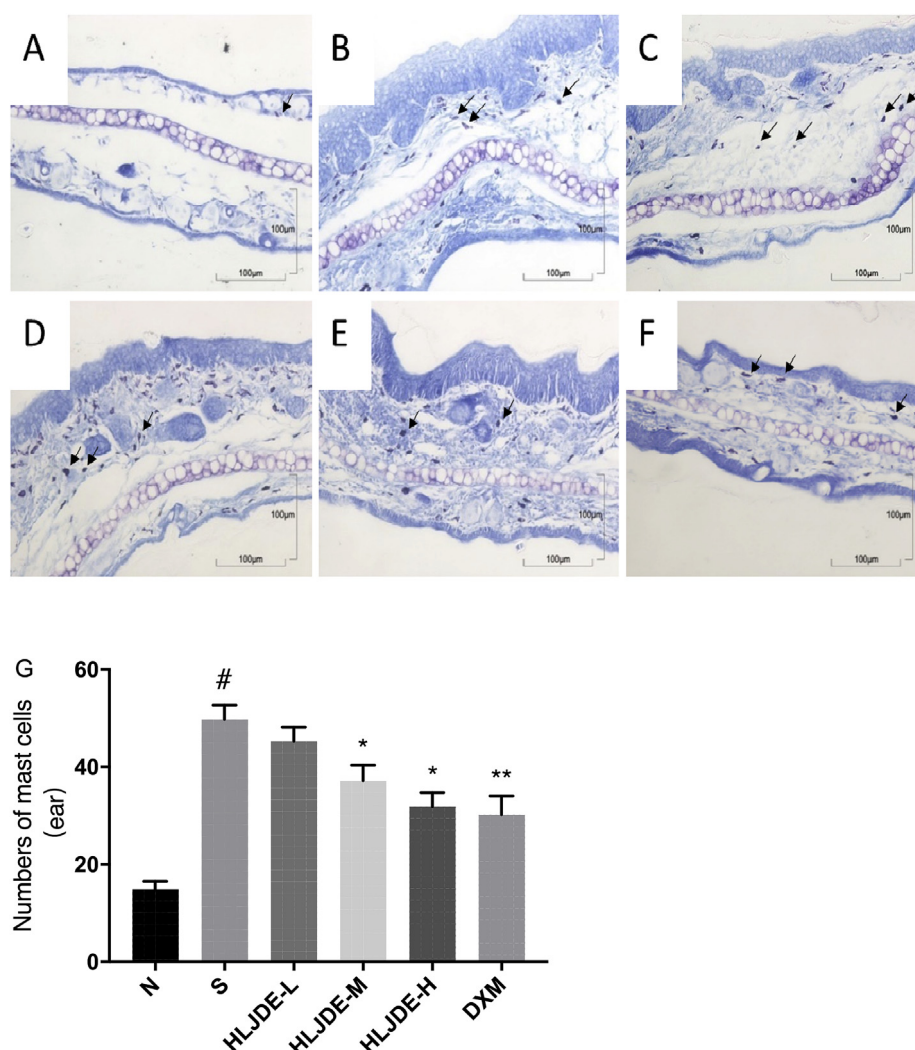


Fig. 8. Effects of HLJDE on the mast cell infiltration in the ears. Ear sections were stained with toluidine blue and analyzed for mast cell infiltration (scale bar = 100 μ m). (A) normal mice, (B) sensitized mice, (C) sensitized mice with DXM, (D–F) sensitized mice with low, medium and high dose of HLJDE, (G) quantification of mast cell infiltration. Data were mean $\pm$ SEM (n = 6). #p < 0.01 vs. normal mice (N); *p < 0.05, **p < 0.01 vs. sensitized mice (S). $\blacktriangledown$: Mast cell. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

3. Results

3.1. Body weight change

As shown in Fig. 2, at the first 3 days before sensitization, the body weights of the mice maintained at approximately 18 g. Sensitization with DNCB caused transient weight loss, but the weight of mice restored to normal level during the interval of sensitization and elicitation phases. However, significant weight loss was observed when the mice were challenged with DNCB. Moreover, treatment with DXM significantly induced additional weight loss (from 18 g to 16 g on day 30) when compared to the DNCB-treated control group. Although a small amount of weight loss by the high dose of HLJDE (from 18 g to 17.5 g) was observed, the mice in this treatment group did not show any behavioral abnormalities.

3.2. Effect of HLJDE on the clinical scores of DNCB-treated mice

The severity of AD-like symptoms was scored upon the completion of the drug treatment. Clinical scores were calculated based on the severity (0, no symptoms; 1, mild; 2, intermediate; 3, severe) of the symptoms of erythema, edema, crusts, excoriations and lichenification. The severity of each symptom was evaluated comparing with the

reference photographs. As shown in Figs. 3–5, at the end of the experiment, severe clinical symptoms, such as excoriation and erythema, were observed in the challenged mice without treatment. Treatments with HLJDE at high and medium dose significantly improved the symptoms on the ear and skin of the mice, especially for the erythema, excoriation, and scaling on skin.

3.3. Effect of HLJDE on epidermal thickness and mast cells in the ears and dorsal skin of DNCB-treated mice

The ears and dorsal skin sections were stained with H & E and toluidine blue to examine infiltration of epidermal thickness and mast cells, respectively. The thickness of epidermis were markedly increased in the ear (Fig. 6) and dorsal skin (Fig. 7) (9.5-fold and 6.3-fold, respectively) in the challenged mice without treatment as compared with the normal control group. In addition, the number of infiltrated mast cells was substantially increased in the ear and (Fig. 8) the dorsal skin (Fig. 9) (233.3% and 450.9%, respectively) in the challenged mice without treatment as compared with the normal control group.

Oral administration of HLJDE significantly attenuated the thickening of epidermis and infiltration of mast cells in both the ear and the dorsal skin of the DNCB-treated mice. In particular, treatment with high dose of HLJDE markedly decreased the epidermal thickness in the ear

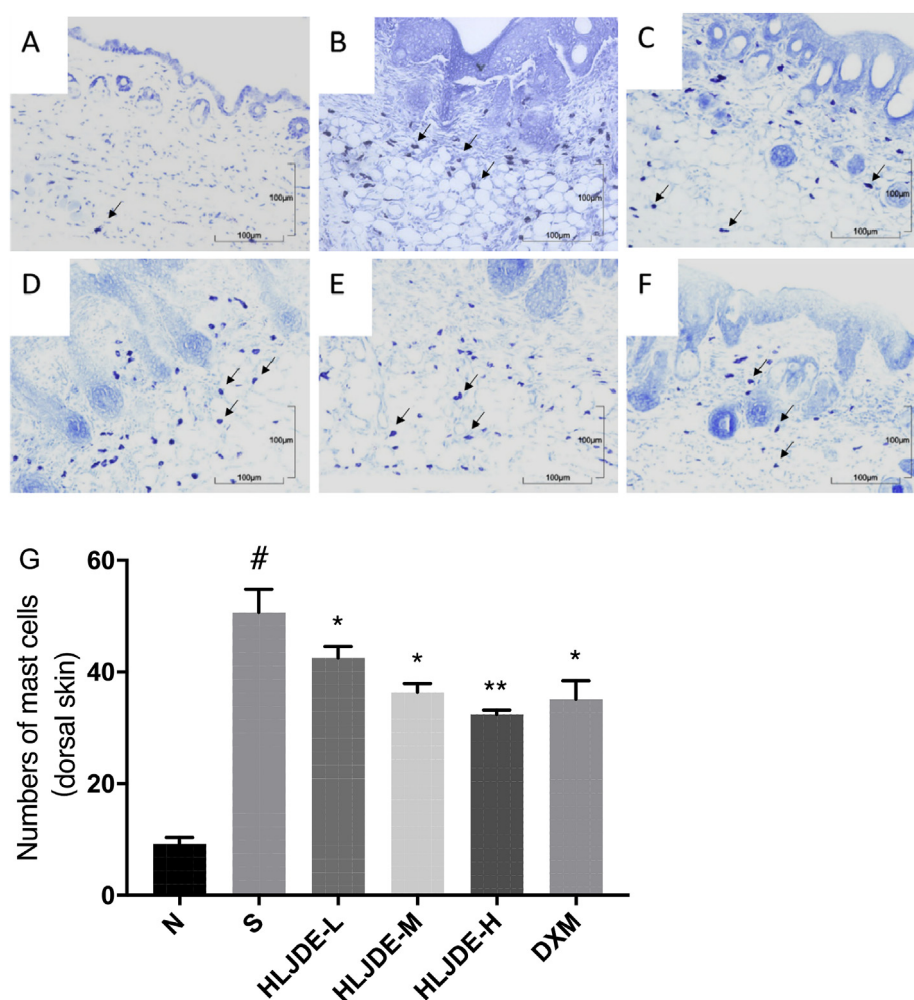


Fig. 9. Effects of HLJDE on the mast cell infiltration in the dorsal skin. Dorsal skin sections were stained with toluidine blue and analyzed for mast infiltration (scale bar = 100 μ m). (A) normal mice, (B) sensitized mice, (C) sensitized mice with DXM, (D–F) sensitized mice with low, medium and high dose of HLJDE, and (G) quantification of mast cell infiltration. Data were mean $\pm$ SEM (n = 6). #p < 0.01 vs. normal mice (N); *p < 0.05, **p < 0.01 vs. sensitized mice (S). $\blacktriangledown$: Mast cell. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

and the skin by 38.2% and 49.3%, respectively, and the therapeutic effects were more potent than that of DXM (Figs. 6–7).

Moreover, HLJDE was able to dose-dependently decrease the number of infiltrated mast cells to 74.7% (medium dose) and 64.0% (high dose) in the ear, and 83.9% (low dose), 71.8% (medium dose) and 63.0% (high dose) in the dorsal skin, as compared with challenged mice group (Figs. 8–9).

3.4. Effects of HLJDE on serum IgE, histamine, IL-4 and TNF- α levels

DNCB sensitization resulted in 31.3 and 4.5-fold increase in the serum IgE and histamine level respectively in BALB/c mice (Fig. 10A–B). HLJDE treatment did not exert significant inhibitory effect on serum IgE and histamine.

Mice sensitized with DNCB saw a nearly 3-fold elevation in serum TNF- α and IL-4 levels (Fig. 10C–D). Treatment with low, medium and high concentration of HLJDE could markedly decrease serum IL-4 by $7.7 \pm 4.2\%$, $29.0 \pm 6.0\%$ and $39.1 \pm 9.5\%$, respectively (Fig. 10C), and suppressed serum TNF- α by $28.2 \pm 6.7\%$, $43.7 \pm 14.5\%$, and $52.2 \pm 11.0\%$, respectively (Fig. 10D). On the other hand, high dose of HLJDE also significantly suppressed the serum TNF- α level in a magnitude similar to that of DXM.

3.5. Effects of HLJDE on the levels of IL-4, TNF- α and TSLP in the dorsal skin

As shown in Fig. 11, the expressions of IL-4, TNF- α and TSLP in the dorsal skin of the sensitized mice accentuated substantially to 874.4 ± 60.1 , 425.7 ± 36.5 and 503.6 ± 42.2 pg/g, respectively. Treatment with HLJDE had significantly inhibitory effect on these cytokines. Treatment with high dose of HLJDE was able to decrease the expression of IL-4, TNF- α and TSLP by $39.6 \pm 13.1\%$, $48.3 \pm 10.1\%$ and $39.4 \pm 10.1\%$, respectively.

3.6. Effects of HLJDE on the filaggrin, MAPKs and NF- κ B pathways

The expression of filaggrin in the dorsal skin was reduced by $38.7 \pm 5.9\%$ upon sensitization with DNCB (Fig. 12). Treatment with the medium dose of HLJDE produced more potent effect than other two doses of HLJDE, and resulted in 1-fold enhancement in the filaggrin expression (Fig. 12B) when compared to model control.

Concerning MAPKs pathways, HLJDE was able to dose-dependently inhibit the activation of p38, JNK and ERK1/2 in the dorsal skin (Fig. 13). Treatment with low, medium and high doses of HLJDE attenuated the p-p38 expression by $52.5 \pm 6.1\%$, $63.8 \pm 8.2\%$ and $57.2 \pm 5.1\%$, respectively (Fig. 13B); reduced the p-JNK expression to $41.9 \pm 7.8\%$, $56.6 \pm 4.8\%$ and $50.1 \pm 4.1\%$ of the control,

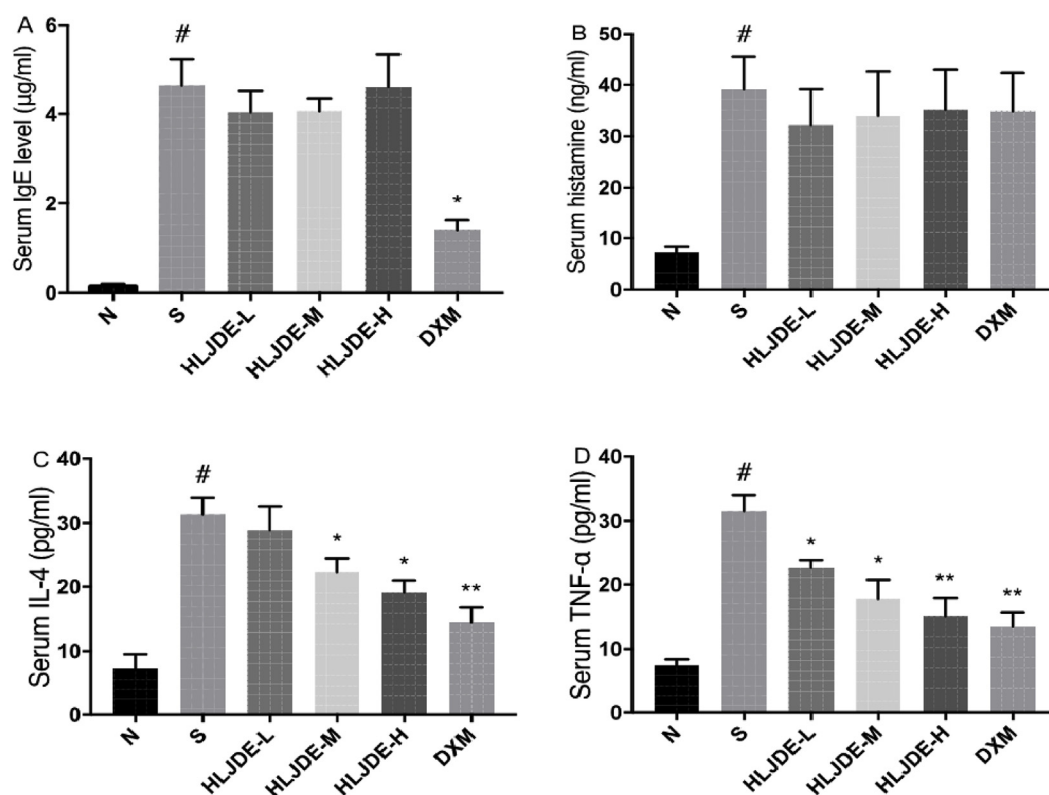


Fig. 10. Effects of HLJDE on the serum IgE (A), histamine (B), IL-4 (C) and TNF-α (D) levels in the sensitized mice. Data were mean $\pm$ SEM (n = 6). [#]p < 0.01 vs. normal mice (N); ^{*}p < 0.05, ^{**}p < 0.01 vs. sensitized mice (S).

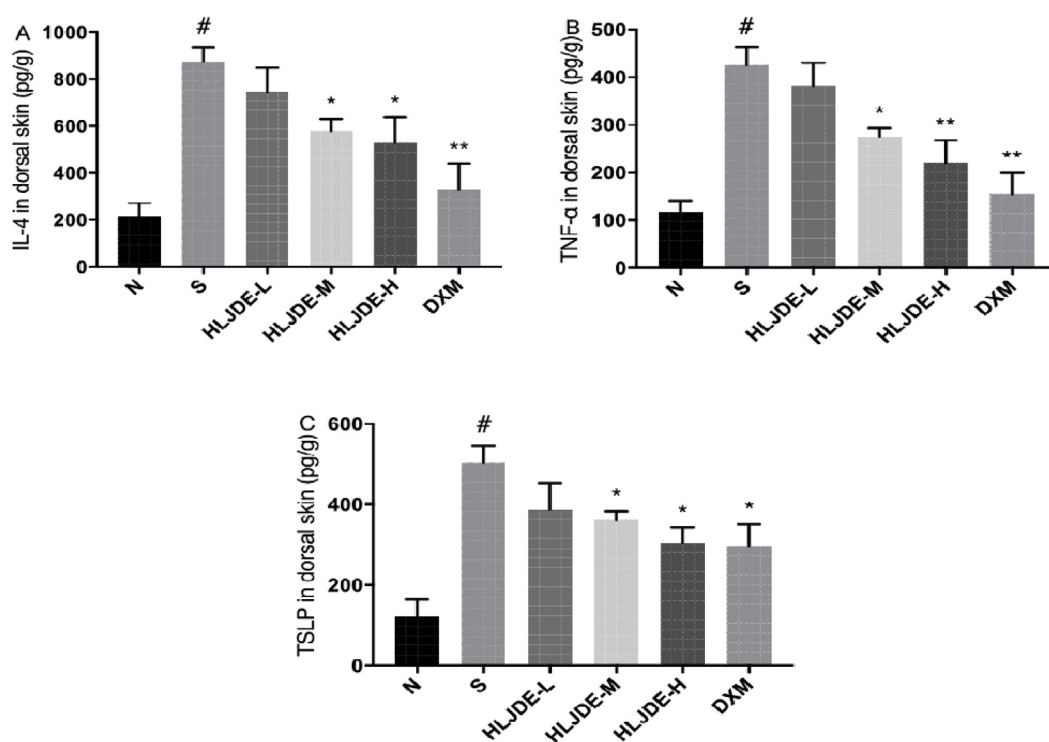


Fig. 11. Effects of HLJDE on the levels of IL-4 (A), TNF-α (B) and TSLP (C) in the dorsal skin of the sensitized mice. Data were mean $\pm$ SEM (n = 6). [#]p < 0.01 vs. normal mice (N); ^{*}p < 0.05, ^{**}p < 0.01 vs. sensitized mice (S).

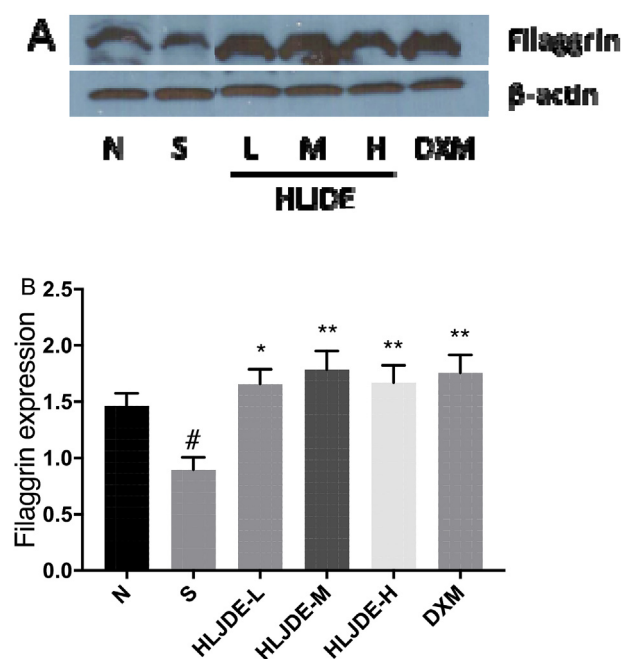


Fig. 12. Effects of HLJDE on the expression of filaggrin in the dorsal skin. Data were mean $\pm$ SEM (n = 6). #p < 0.01 vs. normal mice (N); *p < 0.05, **p < 0.01 vs. sensitized mice (S). (A) Representative Western blot on the expression of filaggrin with β -actin being used as the control. (B) Quantification of filaggrin bands by densitometry.

respectively (Fig. 13C); and decreased the p-ERK expression to $35.6 \pm 5.1\%$, $40.0 \pm 3.1\%$ and $50.1 \pm 3.1\%$, respectively (Fig. 13D).

In addition, HLJDE was capable of dose-dependently increasing the expression of I κ B- α , which was reduced by DNCB sensitization (Fig. 14). In particular, treatment with high dose of HLJDE increased I κ B- α expression by $94.1 \pm 20.5\%$ (Fig. 14B), which was higher than the effect of DXM.

The increased expression of p-p65 was found in the dorsal skin of the sensitized mice. Treatment with HLJDE at low, medium and high dose decreased its expression to $32.5 \pm 1.0\%$, $28.6 \pm 4.1\%$ and $55.7 \pm 6.3\%$, respectively (Fig. 14C). The effect of high dose of HLJDE was similar to that of DXM.

4. Discussion

Our present study amply demonstrated that HLJDE was able to reduce inflammatory symptoms in AD-like mice by suppressing the production of Th2-associated pro-inflammatory cytokines, and inhibiting eosinophil and mast cell infiltration through inactivation of NF- κ B and MAPKs pathways. In addition, HLJDE could also augment the filaggrin expression, which is essential for the restoration of the impaired skin barrier function commonly seen in AD patients.

To validate the therapeutic effect of HLJDE, it is necessary to select appropriate animal model to mimic the human AD. Owing to the complexity of the etiopathogenesis of AD, suitable animal models of this skin disease are still few and far between. There are three main types of animal models for AD, including spontaneous animal models, artificially induced animal models and gene-manipulated animal models, and each model has its own properties and is only suitable for specific research purpose (Jin et al., 2009; Marsella and Olivry, 2003). In view of the time required for model establishment, cost and stability of the model, artificially induced mouse model was found to suit our experiment needs. Repeated topical application of chemical inducer to the same site could induce mouse to develop increased scratching behavior and skin lesions, characterized by thickening of epidermis and dermis and infiltration of eosinophil and mast cells (Jin et al., 2009).

Meanwhile, the serum IgE level and various inflammatory cytokines, such as IL-4, IL-5 and IL-13, were found to be markedly increased on induced mouse model (Chan et al., 2013; HanKang et al., 2014). DNCB is a chemical hapten frequently used to induce AD-like symptoms. In the DNCB-sensitized mice, Th1-dominated inflammatory response is evoked by DNCB and subsequently shift to chronic Th2-dominated inflammatory response after repeated application, which closely resembles AD development in adult patients (Jin et al., 2009). Thus, DNCB-induced Blab/c mouse model is widely adopted to validate therapeutic agents for AD treatment.

The application of DNCB could significantly affect the body weight of mice, and resulted in AD-like skin lesions after several challenges; however, we found no obvious behavioral changes of the sensitized mice albeit weight loss. In our experiment, the severity of AD symptoms in the DNCB-induced mice were significantly improved after treatment with HLJDE, and the results was in agreement with the reduction in the infiltration of eosinophil and mast cells in the ear and dorsal skin. It is believed that the reduced infiltrations of eosinophil and mast cells contribute to a decrease in inflammatory response through decreasing the inflammatory cytokines, thereby the improvement of AD symptoms.

In the AD-like skin lesion, enhanced CD4⁺ activated memory T cell infiltration is involved in immune abnormalities of acute lesion, accompanied with overexpression of Th2 cytokines, such as IL-4, IL-5 and IL-13 (Leung et al., 2004; Leung, 2000). IL-4 plays a critical role in the induction of IgE synthesis and promoting the switch from naïve T cells to allergic type Th2 cells (Vo et al., 2012). In the innate immune responses, the dendritic cells (DCs) can also be activated by the increased TSLP, thereby resulting in infiltration of Th2 cells in the lesional site of AD (Nakajima et al., 2012; Soumelis et al., 2002) and shifting naïve T cells to pro-inflammatory Th2 cells (Liu et al., 2006). Thus, suppressing IL-4 and TSLP level could lead to a decrease in Th2 cells and further to attenuate Th2-dominated inflammatory response. The inhibition on TNF- α is able to reduce immunologic and inflammatory reactions since TNF- α , a key inflammatory mediator, induce other pro-inflammatory cytokines and endothelial adhesion molecules (Chen et al., 2014). In the present study, it is found that HLJDE has potent inhibitory actions on IL-4, TNF- α and TSLP in dorsal skin, even better than DXM except for IL-4. It is likely that IL-4 is more closely associated with IgE response *in vivo* (Turqueti-Neves et al., 2015).

Moreover, the two key inflammatory pathways in lesional skin, including MAPKs and NF- κ B, were suppressed by HLJDE, providing mechanistic explanation on the inhibitory effect on some important inflammatory mediators in the dorsal skin. It is well-known that the activation of NF- κ B pathway by internal/external insults could lead to marked elevation of such inflammatory mediators, thereby the exacerbation of the inflammatory process. In turn, the expression of TNF- α activates NF- κ B again, and forms an amplifying loop (Bonizzi and Karin, 2004). On the other hand, the inhibition of MAPKs could reduce both the synthesis of pro-inflammatory cytokines and their intracellular signaling, and downregulate the activation of NF- κ B (Kaminska et al., 2005; Saklatvala, 2004). Our results unambiguously indicate that HLJDE exerts the anti-inflammatory effect through not only inhibiting the NF- κ B activation, but also blocking a vicious circle involving production of various inflammatory mediators.

Besides local immune abnormalities in lesional skin, AD can also cause the elevation of serum IgE level in patients. It has also been reported that cutaneous sensitization with antigen causes allergic dermatitis and elevated serum IgE level, indicating that the serum IgE might be a key mediator contributing to systemic immune response in AD patients (Leung et al., 2004; Leung, 2000; Ong, 2014). Moreover, in extrinsic AD, IgE-mediated sensitization to allergens plays a key role in the pathogenesis of AD; therefore, inhibition of serum IgE expression is beneficial to attenuate the sensitivity to outside allergens so to improve AD symptoms.

Serum histamine, a key allergic mediator causing inflammatory response, is the major factor contributing to pruritus. In clinic,

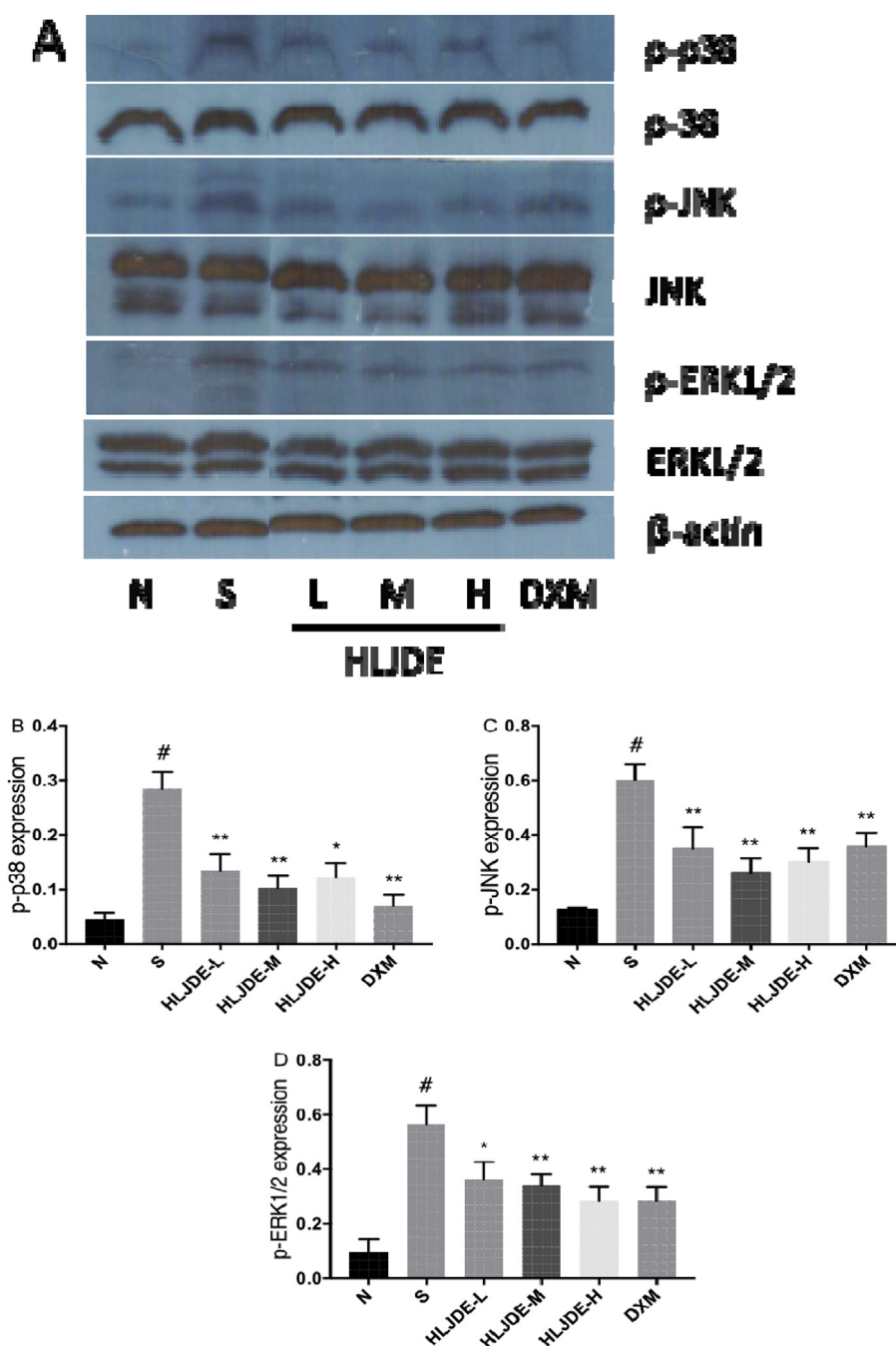


Fig. 13. Effects of HLJDE on MAPKs phosphorylation in the dorsal skin. Data were mean $\pm$ SEM ($n = 6$). # $p < 0.01$ vs. normal mice (N); * $p < 0.05$, ** $p < 0.01$ vs. sensitized mice (S). (A) Representative Western blot on the expression of MAPKs with β -actin being used as the control. (B–D) Quantitative evaluation of p-p38, p-JNK and p-ERK1/2 bands by densitometry.

antihistamines were commonly prescribed to AD patient for alleviating itchiness (Yun et al., 2017). The significant elevation of serum IgE and histamine levels were clearly observed in the DNCB-sensitized mice. However, the distinct differences of actions of HLJDE and DXM are on the regulation of serum IgE and histamine level in the sensitized mice. In our experiment, HLJDE has no effect to reduce the elevated serum IgE and histamine induced by DNCB, suggesting this ancient formula controls the AD symptoms primarily through alleviating skin inflammation in AD-like mice, but not via suppressing IgE-mediated hypersensitivity to extrinsic allergens. Although HLJDE did not exert any effect on serum IgE and histamine level, its inhibitory effects on cytokine production in the serum and dorsal skin, including IL-4, TNF- α and TSLP, lead to good therapeutic effect on the AD-like mice.

HLJDE could not only improve the skin inflammation, but also repair the deranged skin barrier function through enhancing filaggrin expression of lesional skin in the AD-like mouse. Filaggrin is a filament-associated protein in epithelial cell, and plays a key role in skin barrier by preventing water loss and invasion of pathogenic insults through the skin. Experiments on mouse model of filaggrin null mutations (FLG) have shown that filaggrin deficiency in the skin results in an increased penetration of both pathogens and antigens, and induces the elevation of Th2 response and total serum IgE level (Peng and Novak, 2015). Additionally, previous clinical studies have suggested that decreased expression of filaggrin can be observed in AD patients. Furthermore, the correlation between FLG and occurrence of AD in European population has been reported (Palmer et al., 2006; Smith et al., 2006).

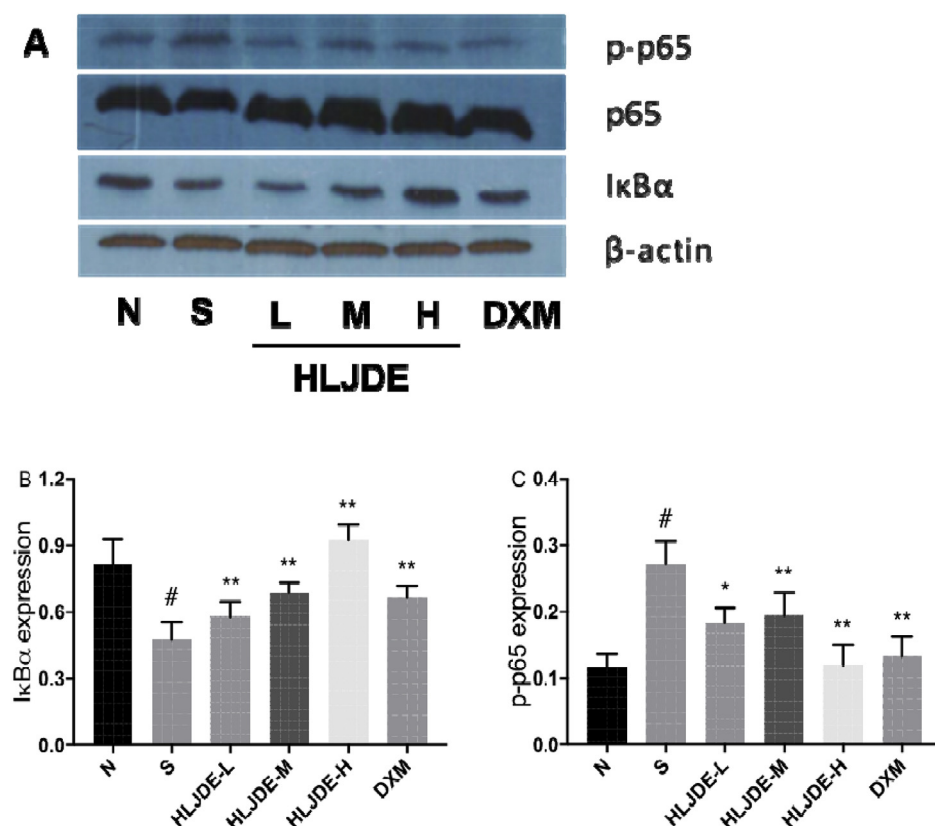


Fig. 14. Effects of HLJDE on IκBα expression and NF-κB phosphorylation in the dorsal skin. Data were mean ± SEM (n = 6). #p < 0.01 vs. normal mice (N); *p < 0.05, **p < 0.01 vs. sensitized mice (S). (A) Representative Western blot on the expression of IκBα with β-actin being used as the control, and the expression of p-p65 with p65 being used as the control. (B–C) Quantitative evaluation of IκBα and p-p65 bands by densitometry.

In our experiments, the decreased expression of filaggrin was found in the sensitized mice without treatment, leading to the impaired skin barrier function. Both HLJDE and DXM were able to elevate filaggrin expression, suggesting their ability to repair the suboptimal skin barrier function in this animal AD model. The repaired skin barrier function contributes to the enhanced defense against the penetration of outside irritants and allergens, thereby blocking the cutaneous inflammation.

5. Conclusions

In summary, our experimental results indicate that HLJDE treatment was able to significantly ameliorate the AD-like symptoms and repair the dysfunctional skin barrier on the DNCB-sensitized mouse. The underlying mechanisms of action involve suppressing the production of inflammatory mediators and enhancing the protein expression of filaggrin through inactivation of MAPKs and NF-κB pathways. The present experimental findings verify the therapeutic effect of HLJDE and help provide scientific rationale for prescribing this herbal formula for AD treatment.

Declarations

Data availability

The data used to support the findings of this study are available from the corresponding author upon request.

Author contributions

LL and LZ conceived and designed this study. CYL performed the *in vivo* studies and collected the data. XYF, LL and LZ analyzed the data. CYL and XYF drafted the manuscript. CWY and LS revised the manuscript. All authors read and approved the final manuscript.

Declaration of competing interest

The authors declare that they have no competing interests.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2019.112367>.

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